设 $F(x,y)=0$ 若 $F(x_0,y_0)=0$，$F_y(x_0,y_0)\neq 0$

$\Longrightarrow \exists$ 隐函数 $y=y(x)$：$\Leftrightarrow F(x,y(x))\equiv 0$

$\Leftrightarrow y(x)-AF(x,y(x))=y(x)$ $\quad A\neq 0$

$\Leftrightarrow Ty(x)\triangleq y(x)-AF(x,y(x))$ 有不动点 $y(x)$：$Ty(x)=y(x)$，

$y_0(x):\triangleq y_0$，$y_1=Ty_0(x)$，$\cdots$ $y_{n+1}(x)=Ty_n(x)$，若 $\underline{|1-AF_y(x,y)|\leq\frac{1}{2}}$

$y_n(x)\rightrightarrows f(x)$：———— 隐函数

推广：$\begin{cases}F_1(x,y,z)=0\\F_2(x,y,z)=0\end{cases}\Rightarrow\begin{cases}y=y(x)\\z=z(x)\end{cases}$：

$Y=\begin{pmatrix}y\\z\end{pmatrix}$：$\vec{F}=\begin{pmatrix}F_1\\F_2\end{pmatrix}$，$\quad A$：$2\times2$ 非退化矩阵，

$\vec{F}(x,Y)=\begin{pmatrix}F_1(x,y,z)\\F_2(x,y,z)\end{pmatrix}=\vec{0}\Leftrightarrow Y-A\vec{F}(x,Y)=Y$

$T: Y\mapsto TY\triangleq Y-A\vec{F}(x,Y)$.

$Y_0(x)=\begin{pmatrix}y_0\\z_0\end{pmatrix}$，$Y_1(x)=TY_0(x)=\begin{pmatrix}y_0\\z_0\end{pmatrix}-A\begin{pmatrix}F_1(x,y_0,z_0)\\F_2(x,y_0,z_0)\end{pmatrix}$

$Y_{n+1}(x)=TY_n(x)=Y_n(x)-A\vec{F}(x,Y_n(x))$ $\qquad Y_n=\begin{pmatrix}y_n\\z_n\end{pmatrix}$

$=\begin{pmatrix}y_n(x)\\z_n(x)\end{pmatrix}-A\begin{pmatrix}F_1(x,y_n(x),z_n(x))\\F_2(x,y_n(x),z_n(x))\end{pmatrix}$

$Y_{n+1}(x)-Y_n(x)=\left(Y_n(x)-A\vec{F}(x,Y_n(x))\right)-\left(Y_{n-1}-A\vec{F}(x,Y_{n-1}(x))\right)$

$=\left(Y_n(x)-Y_{n-1}(x)\right)-A\left(\vec{F}(x,Y_n(x))-\vec{F}(x,Y_{n-1}(x))\right)$.

$$= \begin{pmatrix} y_n(x) - y_{n-1}(x) \\ z_n(x) - z_{n-1}(x) \end{pmatrix} - A \begin{pmatrix} F_1(x, y_n, z_n) - F_1(x, y_{n-1}, z_{n-1}) \\ F_2(x, y_n, z_n) - F_2(x, y_{n-1}, z_{n-1}) \end{pmatrix}$$

$$= (Y_n(x) - Y_{n-1}(x)) - A \begin{pmatrix} F_1(x, \boxed{y_n}, z_n) - F_1(x, \boxed{y_{n-1}}, z_n) + F_1(x, y_{n-1}, \boxed{z_n}) - F_1(x, y_{n-1}, \boxed{z_{n-1}}) \\ F_2(x, y_n, z_n) - F_2(x, y_{n-1}, z_n) + F_2(x, y_{n-1}, z_n) - F_2(x, y_{n-1}, z_{n-1}) \end{pmatrix}$$

$$= (Y_n(x) - Y_{n-1}(x)) - A \begin{pmatrix} F_{1y}(x, \xi_1(x), z_n)(y_n - y_{n-1}) + F_{1z}(x, y_{n-1}, \xi_2(x))(z_n - z_{n-1}) \\ F_{2y}(x, \xi_3(x), z_n)(y_n - y_{n-1}) + F_{2z}(x, y_{n-1}, \xi_4(x))(z_n - z_{n-1}) \end{pmatrix}$$

$$= (Y_n(x) - Y_{n-1}(x)) - A \begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix} \left[\begin{pmatrix} y_n \\ z_n \end{pmatrix} - \begin{pmatrix} y_{n-1} \\ z_{n-1} \end{pmatrix} \right]$$

$Y_n(x) - Y_{n-1}(x)$

$$Y_{n+1}(x) - Y_n(x) = \left[I - \underset{\substack{\uparrow \\ \text{矩阵}}}{A} \begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix} \right] (Y_n(x) - Y_{n-1}(x))$$

$Y_n(x) \overset{?}{\rightrightarrows} Y(x)$ $\qquad \begin{pmatrix} y_n(x) \\ z_n(x) \end{pmatrix} \rightrightarrows \begin{pmatrix} y(x) \\ z(x) \end{pmatrix}$.

? $\|Y_{n+1}(x) - Y_n(x)\| \leq \frac{1}{2} \|Y_n(x) - Y_{n-1}(x)\|$.

$\|\cdot\|$ —— 欧氏模; $\qquad Y = \begin{pmatrix} y \\ z \end{pmatrix}$: $\|Y\| = \sqrt{y^2 + z^2}$;

$B = (b_{ij})_{n \times n}$. $\qquad y = Bx \in \mathbb{R}^n$; $\quad x \in \mathbb{R}^n$,

$\|Bx\| \leq \|B\| \, \|x\|$, $\quad \forall x \in \mathbb{R}^n$. ✓

可以算出来! (留作练习).

B: 矩阵; $\|B\|$ ← 矩阵范数 $\quad \|x\|$: x 的模: 范数

$\because$ Jacobi矩阵 $J = \left.\frac{\partial(F_1, F_2)}{\partial(y, z)}\right|_{(x_0, y_0, z_0)} = \left.\begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix}\right|_{(x_0, y_0, z_0)}$ 非退化.

$\therefore$ 可取 $A = J^{-1}$. $I - AJ = 0$. 即 $I - \underset{\sim}{A}\underbrace{\begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix}}\Big|_{(x_0, y_0, z_0)} = 0$;

F_1, F_2: 连续可微, $\Longrightarrow$ $\exists\ (x_0, y_0, z_0)$ 之邻域 D_1: $\ni$

$$D_1: \begin{cases} |x - x_0| \le \eta_1 & (\le a), \\ |y - y_0| \le b_1 & (\le b), \\ |z - z_0| \le c_1 & (\le c), \end{cases}$$

$B \triangleq I - A\begin{pmatrix} F_{1y}, & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix}$ 在 D_1 中满足:

"$\|B\| \le \frac{1}{2}$". $\forall (x, y, z) \in D_1$ 即 "$\|Bx\| \le \frac{1}{2}\|x\|$. $\forall x \in \mathbb{R}^n$"

$$\Longrightarrow \underline{\|Y_{n+1}(x) - Y_n(x)\|} = \|B(Y_n(x) - Y_{n-1}(x))\|$$
$$\le \frac{1}{2}\|Y_n(x) - Y_{n-1}(x)\|.$$

余下与单个方程相同. #

例 设有方程组 $\begin{cases} xu - yv = 0 \\ yu + xv = 0 \end{cases}$ 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

解: 由题意, 方程组决定隐函数 $\begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$

令 $\begin{cases} F_1 \triangleq xu - yv = 0 \\ F_2 \triangleq yu + xv = 0 \end{cases} \Longrightarrow \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$

Jacobi行列式 $\frac{\partial(F_1, F_2)}{\partial(u, v)} \ne 0$:

由 $\dfrac{\partial(F_1,F_2)}{\partial(u,v)} = \begin{vmatrix} F_{1u} & F_{1v} \\ F_{2u} & F_{2v} \end{vmatrix} = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2+y^2 \neq 0$

$\exists$ 隐函数 $\begin{cases} u=u(x,y) \\ v=v(x,y) \end{cases}$ 且连续可微.

$\begin{cases} F_1 \triangleq xu - yv = 0 \\ F_2 \triangleq yu + xv = 0 \end{cases}$ 两边对 x 求偏导数

$\begin{cases} x\,u_x - y\,v_x + u = 0 \\ y\,u_x + x\,v_x + v = 0 \end{cases}$ 系数行列式 $J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2+y^2 \neq 0$

即 在 $(0,0)$ 以外: $u_x = \dfrac{1}{J}\begin{vmatrix} -u & -y \\ -v & x \end{vmatrix} = \dfrac{1}{J}(-xu - yv)$.

$v_x = \dfrac{1}{J}\begin{vmatrix} x & -u \\ y & -v \end{vmatrix} = \dfrac{1}{J}(yu - xv)$.

同理 可求 u_y, v_y.

$\begin{cases} F_1(x,y,u,v) \triangleq xu - yv = 0 \\ F_2(x,y,u,v) \triangleq yu + xv = 0 \end{cases}$

求微分: $\begin{cases} F_{1x}\,dx + F_{1y}\,dy + F_{1u}\,du + F_{1v}\,dv = 0 \\ F_{2x}\,dx + F_{2y}\,dy + F_{2u}\,du + F_{2v}\,dv = 0 \end{cases}$

解得 $\begin{cases} du = (u_x)\,dx + (u_y)\,dy \\ dv = (v_x)\,dx + (v_y)\,dy \end{cases}$

例 $\begin{cases} x=-u^2+v+z \\ y=u+vz \end{cases}$ 求 $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial z}$.

解: 依题意, x, y, z 是自变量, $u=u(x,y,z)$, $v=v(x,y,z)$ 是隐函数组. 令

$$\begin{cases} F_1 = x+u^2-v-z \\ F_2 = y-u-vz. \end{cases}$$

$\begin{cases} F_1=0 = x+u^2-v-z \\ F_2=0 = y-u-vz \end{cases}$ 决定隐函数之条件.

当 $\frac{\partial(F_1,F_2)}{\partial(u,v)} = \begin{vmatrix} 2u & -1 \\ -1 & -v \end{vmatrix} = -2uv-1 \neq 0$ 时

即: $2uv+1\neq 0$ 时. ∃隐函数组: $\begin{cases} u=u(x,y,z) \\ v=v(x,y,z) \end{cases}$

$\begin{matrix} dF_1=0 \\ dF_2=0 \end{matrix} \Longrightarrow$ 求 du. dv. $\Longrightarrow \frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial z}$.

#.

§16.2. 函数行列式之性质. 坐标变换.

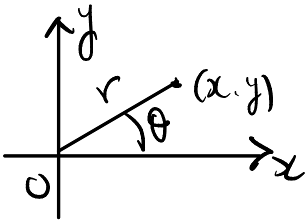
设一元函数 $y=f(x)$: $x\in[a,b]$. $f'(x)\neq 0$: 若 $f'(x)>0$:

$\Longrightarrow f(x)$ 严格↗: $\Longrightarrow$ ∃反函数 $x=g(y)$. 可知

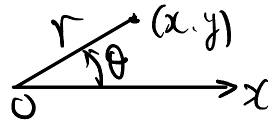
$f(g(y))\equiv y$; $g(f(x))\equiv x$.

$f'(x)\cdot g'(y)=1 \Rightarrow g'(y)=\dfrac{1}{f'(x)}$.

二维：直角坐标与极坐标变换：



$$\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}\Leftrightarrow\begin{cases}r=\sqrt{x^2+y^2}\\ \theta=\arctan\dfrac{y}{x}\end{cases}$$



一. 复合函数的Jacobi行列式：

设 $\begin{cases}u=F_1(x,y)\\ v=F_2(x,y)\end{cases}$ $\begin{cases}x=\varphi(s,t)\\ y=\psi(s,t)\end{cases}$ 连续可微，且可以

~~复合成~~ 复合函数：$\begin{cases}u=F_1(\varphi(s,t),\psi(s,t))\\ v=F_2(\varphi(s,t),\psi(s,t))\end{cases}$

$$\frac{\partial(F_1,F_2)}{\partial(s,t)}=\begin{vmatrix}F_{1s} & F_{1t}\\ F_{2s} & F_{2t}\end{vmatrix}$$

$$=\det\begin{pmatrix}F_{1x}\cdot\varphi_s+F_{1y}\cdot\psi_s, & F_{1x}\cdot\varphi_t+F_{1y}\cdot\psi_t\\ F_{2x}\cdot\varphi_s+F_{2y}\cdot\psi_s, & F_{2x}\cdot\varphi_t+F_{2y}\cdot\psi_t\end{pmatrix}$$

矩阵的行列式

$$=\det\left(\begin{pmatrix}F_{1x} & F_{1y}\\ F_{2x} & F_{2y}\end{pmatrix}\begin{pmatrix}\varphi_s & \varphi_t\\ \psi_s & \psi_t\end{pmatrix}\right)$$

$$=\begin{vmatrix}F_{1x} & F_{1y}\\ F_{2x} & F_{2y}\end{vmatrix}\cdot\begin{vmatrix}\varphi_s & \varphi_t\\ \psi_s & \psi_t\end{vmatrix}$$

$$= \frac{\partial(F_1, F_2)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(s, t)} .$$ ← Chain rule.

"链式法则"

$f(x)$, $x = g(t)$ 可微,

$$\Rightarrow \frac{d}{dt} f(g(t)) = f'(x) \cdot g'(t).$$ Chain Rule

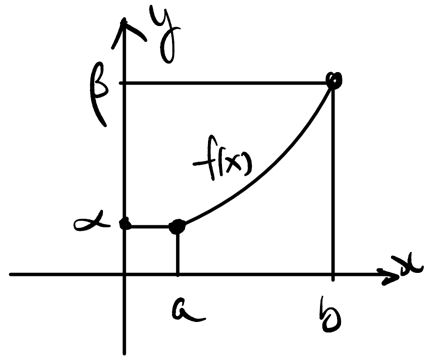
更一般地:

定理2.1 设 $y_i = f_i(x)$ $(i = 1, 2, \cdots, n)$ 是 $x = (x_1, x_2, \cdots, x_n) \in D$ $(D \subset \mathbb{R}^n)$ 连续可微, $x_i = \varphi_i(t)$ 是 $t = (t_1, t_2, \cdots, t_n) \in \Omega$ 连续可微, 且 $\forall t \in \Omega$, $(\varphi_1(t), \varphi_2(t), \cdots, \varphi_n(t)) \in D$,

复合函数 $y_i = f_i(\varphi_1(t), \varphi_2(t), \cdots, \varphi_n(t))$ 满足:

$$\frac{\partial(f_1, f_2, \cdots, f_n)}{(t_1, t_2, \cdots, t_n)} = \frac{\partial(f_1, f_2, \cdots, f_n)}{\partial(x_1, x_2, \cdots, x_n)} \cdot \frac{\partial(x_1, x_2, \cdots, x_n)}{\partial(t_1, t_2, \cdots, t_n)}.$$

二、坐标变换:



一元函数 $f(x) \in C^1[a, b]$

若 $f'(x) > 0$: 严格单调, $\alpha = f(a)$, $\beta = f(b)$.

$f(x)$ 是一个坐标变换: $[a, b] \to [\alpha, \beta]$

推广到 n 维情形:

定理2. 设 $y_i = f_i(x)$ 是 $x \in D \subset \mathbb{R}^n$ 的连续可微函数

$x=(x_1, x_2, \cdots, x_n)\in D$. 如果 Jacobi 行列式 $\frac{\partial(f_1, f_2, \cdots, f_n)}{\partial(x_1, x_2, \cdots, x_n)}\neq 0$

则 ∃ 逆变换 $x_i=\varphi_i(y)$ $(1\leq i\leq n)$:

$$y_i\equiv f_i(\varphi_1(y), \varphi_2(y), \cdots, \varphi_n(y))$$

$$x_i\equiv \varphi_i(f_1(x), f_2(x), \cdots, f_n(x))$$

且: $\left(\frac{\partial f_i}{\partial x_j}\right)_{n\times n}\cdot\left(\frac{\partial \varphi_i}{\partial y_j}\right)_{n\times n}=I$.

证明: 令 $F_i(x,y)=f_i(x_1, x_2, \cdots, x_n)-y_i=0$ $1\leq i\leq n$.

若 $\frac{\partial(F_1, F_2, \cdots, F_n)}{\partial(x_1, x_2, \cdots, x_n)}\neq 0 \Rightarrow$ 存在唯一 $x_i=\varphi_i(y)$,

$$\frac{\partial(F_1, F_2, \cdots, F_n)}{\partial(x_1, x_2, \cdots, x_n)}=\begin{vmatrix} F_{1x_1} & F_{1x_2} & \cdots & F_{1x_n} \\ F_{2x_1} & F_{2x_2} & \cdots & F_{2x_n} \\ \vdots & \vdots & & \vdots \\ F_{nx_1} & F_{nx_2} & \cdots & F_{nx_n}\end{vmatrix}$$

$$=\begin{vmatrix} f_{1x_1} & f_{1x_2} & \cdots & f_{1x_n} \\ f_{2x_1} & f_{2x_2} & \cdots & f_{2x_n} \\ \vdots & \vdots & & \vdots \\ f_{nx_1} & f_{nx_2} & \cdots & f_{nx_n}\end{vmatrix}$$

$$=\frac{\partial(f_1, f_2, \cdots, f_n)}{\partial(x_1, x_2, \cdots, x_n)}$$

$\Rightarrow \exists!$ 隐函数 $x_i=\varphi_i(y)$, $1\le i\le n$. 带入:

$$F_i(x,y)=f_i(\varphi_1(y),\varphi_2(y),\dots,\varphi_n(y))-y_i=0$$

$$\Rightarrow \quad f_i(\varphi_1(y),\varphi_2(y),\dots,\varphi_n(y))\equiv y_i$$

$x_i=\varphi_i(x)$ 反函数组 #

例. 直角坐标与极坐标:

$$\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases};$$

由 $$\frac{\partial(x,y)}{\partial(r,\theta)}=\begin{vmatrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{vmatrix}=r\cos^2\theta+r\sin^2\theta=r\neq 0$$

$\Rightarrow \exists!$ 反函数组. $\begin{cases} r=\sqrt{x^2+y^2} \\ \theta=\arctan\dfrac{y}{x} \end{cases}$